

A Quantum-Inspired Annealing-Based Algorithm to solve Large-Scale Binary Optimization Problems

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Combinatorial optimization arises throughout science and industry, yet finding good solutions becomes difficult as problem sizes grow and more complex. Many of these combinatorial problems (COPs), usually defined on graphs, can be written as quadratic unconstrained binary optimization (QUBO) hamiltonians, giving a common starting point for optimal configuration search. We introduce the quantum-enhanced free-energy machine (QeFEM), a classical, quantum-inspired method that combines ideas from statistical physics, variational optimization, and annealing. It seeks to find the probabilistic thermal distribution of the graph vertices by mini Gradient-based updates then guide many candidate solutions in parallel toward feasible binary assignments. We derive the method and analyze its search dynamics, then evaluate it on MaxCut and planted Ising problems with different sizes and structures. Across these demanding benchmarks, QeFEM finds high-quality solutions with a common solver framework. Recorded inputs, seeds, and settings support repeatable experiments and controlled comparisons across CPUs and accelerators.

I. INTRODUCTION

Binary optimization asks for an assignment of finitely many decisions that minimizes a cost or maximizes a reward. The variables may describe the two sides of a graph cut, selected facilities, accepted constraints, or a spin configuration. A common mathematical representation is quadratic unconstrained binary optimization (QUBO); after an affine change of variables, the same objective is an Ising energy. The representation is general enough to encode many problems, but it does not remove their combinatorial difficulty: N unrestricted binary variables have 2^N possible assignments. MaxCut and general Ising optimization already contain NP-hard cases [1–3].

For MaxCut, the input is an undirected weighted graph $G = (V, E, w)$ and a feasible solution is a partition $S \subseteq V$ of its vertices. The problem is to find

$$\max_{S \subseteq V} C(S), \quad C(S) = \sum_{\{i,j\} \in E} w_{ij} \mathbf{1}\{|\{i,j\} \cap S| = 1\}. \quad (1)$$

Equivalently, a binary spin $s_i \in \{-1, +1\}$ marks each vertex’s side of the partition. In this paper, a reported MaxCut solution is such a binary partition, scored with the original edge weights; the continuous QeFEM state is the means of finding it.

Many useful solvers therefore exchange exhaustive enumeration for a search over an easier auxiliary space. Simulated annealing explores binary states stochastically [4]. The free-energy machine (FEM) optimizes a factorized thermal distribution with gradients [5]. Local quantum annealing (LQA) instead optimizes a pure product-state surrogate for a changing transverse-field Hamiltonian [6]. Discrete simulated bifurcation (dSB) updates classical position and momentum variables with a sign-dependent interaction [7]. These algorithms differ in objective, dynamics, and computational budget. Their common feature is that a continuous or population-based search eventually returns an ordinary binary assignment.

QeFEM places the search in a product of single-qubit mixed states. A local state has a longitudinal component, which records the expected binary decision; a transverse component, which couples to a driver; and an entropy, which resists premature polarization. The objective is the mean-field free energy of a transverse-field Ising model. We minimize it with classical tensor operations and independently initialized replicas. The word “quantum” identifies the state family and the noncommuting driver in the surrogate Hamiltonian. It does not imply entanglement, physical quantum evolution, or quantum computational speedup.

The point of a detailed derivation is to identify precisely what this relaxation can and cannot promise. The exact Gibbs state is generally intractable, so the product ansatz introduces a substantial approximation. The resulting free energy is nonconvex at low temperature. Annealing and replica search can find strong states, but neither follows an adiabatic theorem, and neither guarantees the optimal cut. Furthermore, the later G-set experiments use a deliberately modified, discrete-neighbour search direction. That update shares the state representation and final score with the smooth method, but it is not the exact gradient of the original free energy. Treating the two variants as identical would obscure both the mathematics and the evidence.

The results require similar care. The main saved comparison consists of frozen QeFEM and LQA configurations on K2000, Wishart, and Chook instances. The G-set result combines historical attainment with a subsequent adaptive search and fresh-seed confirmation. A separate FEM notebook contains informative K2000 runs of FEM and dSB, but those runs are not paired with the saved QeFEM experiment in compute, numerical platform, or seed protocol. We preserve these distinctions throughout rather than compressing unlike measurements into one apparent league table.

This expanded manuscript is organized around four questions. First, how does a binary Ising objective lead to

the QeFEM free energy? Second, what do its stationary equations, gradients, and instability scale say about the search dynamics? Third, what was actually run, including optimizer settings, replica counts, seeds, and readout? Fourth, what do the stored endpoints establish, and where does the method still fall short? The figures are chosen to answer these questions directly: a winner trajectory and entropy trace expose the annealing dynamics; population scaling diagnoses the K2000 plateau; instance-level curves show where planted error changes; and a gap-before/after plot displays which revisited G-set graphs improved and which supplied references were newly attained.

II. FROM BINARY OBJECTIVES TO AN ANNEALED VARIATIONAL PROBLEM

A. QUBO, Ising variables, and MaxCut

Let $\mathbf{x} \in \{0, 1\}^N$ and let $Q = Q^\top$ be a real matrix. We write a QUBO as

$$E_Q(\mathbf{x}) = \mathbf{x}^\top Q \mathbf{x} + c. \quad (2)$$

The symmetric convention is explicit: off-diagonal entries of Q occur twice in the matrix product. Define $s_i = 1 - 2x_i$, so that $x_i = (1 - s_i)/2$ and $s_i \in \{-1, +1\}$. Direct substitution gives

$$\begin{aligned} E_Q(\mathbf{s}) &= c + \frac{1}{4}(\mathbf{1} - \mathbf{s})^\top Q (\mathbf{1} - \mathbf{s}) \\ &= c + \frac{1}{4}\mathbf{1}^\top Q \mathbf{1} - \frac{1}{2}(Q\mathbf{1})^\top \mathbf{s} + \frac{1}{4}\mathbf{s}^\top Q \mathbf{s}. \end{aligned} \quad (3)$$

Since $s_i^2 = 1$, the diagonal part of the last term is constant. With each unordered pair counted once,

$$\begin{aligned} E_I(\mathbf{s}) &= c_0 + \sum_i h_i s_i + \sum_{i < j} K_{ij} s_i s_j, \\ c_0 &= c + \frac{1}{4}\mathbf{1}^\top Q \mathbf{1} + \frac{1}{4} \text{Tr } Q, \\ h_i &= -\frac{1}{2}(Q\mathbf{1})_i, \quad K_{ij} = \frac{1}{2}Q_{ij} \quad (i \neq j). \end{aligned} \quad (4)$$

The choice $s_i = 2x_i - 1$ reverses the sign of h_i but is otherwise equivalent. Writing the affine map matters when results from two software packages are compared; a silent sign or factor-of-two change can alter a reported energy while leaving the spin vector unchanged.

For the MaxCut problem in Eq. (1), encode a partition by $s_i = +1$ on S and $s_i = -1$ otherwise. The edge indicator then becomes $(1 - s_i s_j)/2$, giving

$$C(\mathbf{s}) = \sum_{\{i,j\} \in E} w_{ij} \frac{1 - s_i s_j}{2} = \frac{W - E_G(\mathbf{s})}{2}, \quad (6)$$

where $W = \sum_{\{i,j\} \in E} w_{ij}$ and $E_G(\mathbf{s}) = \sum_{\{i,j\} \in E} w_{ij} s_i s_j$. This identity holds for signed weights as well as nonnegative ones. We use E_G as the Ising energy to be minimized in the graph experiments. If K is its symmetric zero-diagonal coupling matrix, then $E_G(\mathbf{s}) = \frac{1}{2}\mathbf{s}^\top K \mathbf{s}$. For

K2000, the final cut is always recomputed with the original unscaled edge weights, not inferred from the value of the relaxed objective.

B. A transverse-field Gibbs target

Replace each classical spin s_i by the Pauli operator Z_i to form the diagonal problem Hamiltonian

$$H_P = c_0 I + \sum_i h_i Z_i + \sum_{i < j} K_{ij} Z_i Z_j. \quad (7)$$

Its computational-basis eigenvalues are precisely the Ising costs. We introduce a transverse driver with strength $\Gamma \geq 0$,

$$H_\Gamma = H_P - \Gamma \sum_i X_i. \quad (8)$$

At a temperature $T > 0$, the corresponding exact Gibbs state is

$$\sigma_{T,\Gamma} = \frac{e^{-H_\Gamma/T}}{Z_{T,\Gamma}}, \quad Z_{T,\Gamma} = \text{Tr } e^{-H_\Gamma/T}. \quad (9)$$

The state has a $2^N \times 2^N$ matrix representation and is never formed by the algorithm. Its role is variational: it identifies an objective whose terms have a controlled thermodynamic origin. The schedule changes this objective over optimization steps. It is not a physical time evolution of Eq. (8).

For any density matrix ρ , the quantum relative entropy is $D_{\text{KL}}(\rho \parallel \sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)]$. Since $\log \sigma_{T,\Gamma} = -H_\Gamma/T - \log Z_{T,\Gamma}$,

$$\begin{aligned} TD_{\text{KL}}(\rho \parallel \sigma_{T,\Gamma}) &= \text{Tr}(\rho H_\Gamma) - TS(\rho) + T \log Z_{T,\Gamma} \\ &= \mathcal{F}_{T,\Gamma}(\rho) - F_{T,\Gamma}^*, \end{aligned} \quad (10)$$

where $S(\rho) = -\text{Tr}(\rho \log \rho)$, $\mathcal{F}_{T,\Gamma}(\rho) = \text{Tr}(\rho H_\Gamma) - TS(\rho)$, and $F_{T,\Gamma}^* = -T \log Z_{T,\Gamma}$. Positivity of relative entropy implies $\mathcal{F}_{T,\Gamma}(\rho) \geq F_{T,\Gamma}^*$, with equality at the exact Gibbs state. Restricting ρ to an inexpensive family converts this identity into a tractable variational approximation [8, 9]. It does not make the restricted minimum equal to the exact free energy.

C. Product states and the Bloch disk

QeFEM restricts the trial state to

$$\begin{aligned} \rho &= \bigotimes_{i=1}^N \rho_i, \\ \rho_i &= \frac{1}{2}(I + m_{x,i} X_i + m_{z,i} Z_i), \\ m_{x,i}^2 + m_{z,i}^2 &\leq 1. \end{aligned} \quad (11)$$

No intersite entanglement or classical correlation is represented in this family. The restriction to the x - z Bloch

disk is sufficient for minimizing the product-state free energy of the real Hamiltonian Eq. (8). Indeed, H_Γ contains no Y_i expectation, while the local entropy decreases as the Bloch length increases. Given any site with a nonzero y component, setting that component to zero preserves the energy and weakly increases the entropy. It therefore cannot increase $\mathcal{F}_{T,\Gamma}$. The two-coordinate ansatz is a consequence of the chosen Hamiltonian, not an arbitrary truncation of a needed y degree of freedom.

Let $R_i = \sqrt{m_{x,i}^2 + m_{z,i}^2}$. The eigenvalues of ρ_i are $(1 \pm R_i)/2$, so its entropy is the binary entropy of those eigenvalues:

$$S_i = -\frac{1+R_i}{2} \log \frac{1+R_i}{2} - \frac{1-R_i}{2} \log \frac{1-R_i}{2}. \quad (12)$$

The full entropy is additive, $S(\rho) = \sum_i S_i$. Its maximum is $N \log 2$, attained at the maximally mixed product state. In contrast, $S_i \rightarrow 0$ as $R_i \rightarrow 1$.

For a product state, $\langle Z_i \rangle = m_{z,i}$ and $\langle Z_i Z_j \rangle = m_{z,i} m_{z,j}$ for $i \neq j$. Consequently the restricted free energy is

$$\boxed{\mathcal{F}_{T,\Gamma}(\mathbf{m}_x, \mathbf{m}_z) = c_0 + \sum_i h_i m_{z,i} + \sum_{i < j} K_{ij} m_{z,i} m_{z,j} - \Gamma \sum_i m_{x,i} - T \sum_i S_i.} \quad (13)$$

This is the smooth QeFEM objective. The first line is the exact energy expectation *within the product ansatz*; factorization of pair expectations is the approximation to the exact Gibbs state. The transverse term rewards positive x polarization, and the entropy term favors local mixture at high temperature. The constant c_0 affects reported energies but not the variational minimizer.

The longitudinal component also has a direct probabilistic interpretation. A Z_i measurement returns $s_i \in \{-1, +1\}$ with probability

$$q_i(s_i) = \frac{1 + s_i m_{z,i}}{2}, \quad q(\mathbf{s}) = \prod_i q_i(s_i). \quad (14)$$

This distribution is a useful interpretation of the state; the benchmark solver does not sample it for final scoring. It uses deterministic sign readout after optimization, as specified in Sec. IV.

III. STATIONARY STRUCTURE AND ANNEALING DYNAMICS

A. Unconstrained fields and exact local derivatives

Directly optimizing (m_x, m_z) would require projection back into the Bloch disk. Instead, each site has unconstrained real fields $\mathbf{a}_i = (u_i, v_i)^\top$ and radius $r_i =$

$\sqrt{u_i^2 + v_i^2}$. Define

$$\mathbf{m}_i = (m_{x,i}, m_{z,i})^\top = g(r_i) \mathbf{a}_i, \quad g(r) = \begin{cases} \tanh r/r, & r > 0, \\ 1, & r = 0. \end{cases} \quad (15)$$

Then $R_i = \tanh r_i < 1$ for finite fields, so positivity of every local density matrix is automatic. More explicitly,

$$\rho_i = \frac{e^{u_i X_i + v_i Z_i}}{2 \cosh r_i}, \quad (16)$$

because $(u_i X_i + v_i Z_i)^2 = r_i^2 I$. Equation (15) is therefore an exponential-family parameterization of the open Bloch disk. At $r_i = 0$, it is interpreted by continuity rather than evaluated as $0/0$.

The local entropy is especially stable in these coordinates:

$$S_i = \log(2 \cosh r_i) - r_i \tanh r_i. \quad (17)$$

The implementation evaluates the log-cosh term by a stable log-sum-exp operation to avoid overflow. Differentiating Eq. (15) gives the symmetric Jacobian

$$J_i = \frac{\partial \mathbf{m}_i}{\partial \mathbf{a}_i} = g(r_i) I + \frac{g'(r_i)}{r_i} \mathbf{a}_i \mathbf{a}_i^\top, \quad g'(r) = \frac{r \operatorname{sech}^2 r - \tanh r}{r^2}. \quad (18)$$

Its tangential eigenvalue is $\tanh r/r$, and its radial eigenvalue is $\operatorname{sech}^2 r$. Both are positive at finite r ; both tend to one as $r \rightarrow 0$. The radial eigenvalue becomes small when the local state is nearly pure, an explicit reason that field-space optimization can slow near the boundary even while the physical-coordinate gradient remains large.

Define the instantaneous longitudinal effective field

$$H_i(\mathbf{m}_z) = h_i + \sum_{j \neq i} K_{ij} m_{z,j}. \quad (19)$$

Because $dS_i/dR_i = -\operatorname{arctanh} R_i$,

$$\frac{\partial \mathcal{F}}{\partial m_{x,i}} = -\Gamma + T \frac{\operatorname{arctanh} R_i}{R_i} m_{x,i}, \quad (20)$$

$$\frac{\partial \mathcal{F}}{\partial m_{z,i}} = H_i + T \frac{\operatorname{arctanh} R_i}{R_i} m_{z,i}. \quad (21)$$

The quotient tends to one as $R_i \rightarrow 0$. Since $\operatorname{arctanh} R_i = r_i$ and $\mathbf{m}_i/R_i = \mathbf{a}_i/r_i$, the two physical-coordinate gradients form the remarkably simple vector

$$\nabla_{\mathbf{m}_i} \mathcal{F} = \begin{pmatrix} -\Gamma \\ H_i \end{pmatrix} + T \mathbf{a}_i. \quad (22)$$

Thus the exact gradient seen by the optimizer is

$$\boxed{\nabla_{\mathbf{a}_i} \mathcal{F} = J_i \begin{pmatrix} -\Gamma \\ H_i \end{pmatrix} + T \operatorname{sech}^2(r_i) \mathbf{a}_i.} \quad (23)$$

The second term follows from $J_i \mathbf{a}_i = \text{sech}^2(r_i) \mathbf{a}_i$. This expression agrees with automatic differentiation of Eq. (13); Appendix A shows the chain rule in detail. Exact manual evaluation is an implementation choice, whereas changing H_i changes the vector field being optimized.

B. Self-consistency and the classical endpoint

At any finite r_i , J_i is invertible. A stationary point therefore satisfies $\nabla_{\mathbf{m}_i} \mathcal{F} = 0$, giving

$$u_i = \frac{\Gamma}{T}, \quad v_i = -\frac{H_i(\mathbf{m}_z)}{T}. \quad (24)$$

Substitution into the Bloch map yields the product-state self-consistency equation

$$m_{z,i} = -\frac{H_i}{\sqrt{\Gamma^2 + H_i^2}} \tanh\left(\frac{\sqrt{\Gamma^2 + H_i^2}}{T}\right), \quad (25)$$

with the continuous zero-field limit understood. The transverse component is

$$m_{x,i} = \frac{\Gamma}{\sqrt{\Gamma^2 + H_i^2}} \tanh\left(\frac{\sqrt{\Gamma^2 + H_i^2}}{T}\right). \quad (26)$$

These are coupled equations: changing one site's m_z changes the fields of its neighbors. Their multiple solutions at low temperature are one manifestation of the non-convex landscape. The optimizer does not solve Eq. (25) by fixed-point iteration; its annealed gradient updates need not be at equilibrium at any intermediate step.

If $\Gamma = 0$ while $T > 0$, Eq. (25) reduces to the classical mean-field equation

$$m_{z,i} = -\tanh\left(\frac{H_i}{T}\right), \quad m_{x,i} = 0. \quad (27)$$

At $T \rightarrow 0$ and $\Gamma \rightarrow 0$, strongly nonzero local fields drive $|m_z| \rightarrow 1$. This limiting statement is local and conditional. Finite optimizer runs can retain uncertain sites or become trapped in suboptimal sign patterns. If $T \rightarrow 0$ at fixed $\Gamma > 0$, the transverse component need not disappear; removing the driver is essential to the intended classical endpoint.

Proposition 1 (Relation to the classical FEM family). *At $\Gamma = 0$, the minimum of Eq. (13) over the product-state Bloch disks equals its minimum over the subfamily $m_{x,i} = 0$ for every site. Hence the transverse state coordinate does not improve the optimal static classical mean-field free energy at the endpoint.*

Proof. For fixed $m_{z,i}$, the problem energy is independent of $m_{x,i}$. Setting $m_{x,i} = 0$ decreases R_i and weakly increases the entropy S_i , because S_i is decreasing in R_i . At $\Gamma = 0$, the driver term vanishes. The change therefore weakly decreases the free energy at every site. Applying it to an arbitrary product state proves the claim. $\square$

This fact resolves an apparent paradox in method comparisons. A transverse coordinate can change the *path* through which a basin is entered, but it does not imply a theorem that QeFEM must defeat an optimized classical FEM run on every instance. The practical algorithms also use different parameterizations, schedules, normalizations, and optimizers. Their endpoint quality is an empirical question.

C. A curvature scale for the onset of polarization

The zero-field case $h_i = 0$ admits a particularly informative local analysis. The unpolarized longitudinal state $\mathbf{m}_z = 0$ has $m_x = \tanh(\Gamma/T)$ at a stationary point. Let $\lambda_{\min}(K)$ be the smallest eigenvalue of the symmetric coupling matrix, and write $\kappa = -\lambda_{\min}(K)$ when this eigenvalue is negative. The longitudinal Hessian at that state is

$$\nabla_{\mathbf{m}_z}^2 \mathcal{F}|_{\mathbf{m}_z=0} = K + q(T, \Gamma)I, \quad q(T, \Gamma) = \frac{\Gamma}{\tanh(\Gamma/T)}, \quad (28)$$

where $q(T, 0) = T$ by continuity. To see the local term, use $\text{arctanh } R/R$ in Eq. (21) at $R = \tanh(\Gamma/T)$: it becomes $(\Gamma/T)/\tanh(\Gamma/T)$ and is multiplied by T . The first unstable longitudinal mode appears when $q(T, \Gamma)$ falls through κ . If $q > \kappa$, all longitudinal eigenmodes have positive curvature at this stationary state; if $q < \kappa$, at least one has negative curvature.

This criterion explains the later G-set spectral schedules. They parameterize a desired q relative to the graph-specific κ , rather than cooling solely according to an absolute temperature that has a different meaning on each weighted graph. Defining $\rho = \Gamma/T$, one can recover physical schedule values from q and ρ :

$$T = q \frac{\tanh \rho}{\rho}, \quad \Gamma = q \tanh \rho, \quad (29)$$

with $T = q$ when $\rho = 0$. A schedule that lowers q/κ from above one to below one moves through the local polarization threshold. This is a statement about curvature around one stationary state, not a proof that the ensuing trajectory finds the ground state.

At sufficiently high temperature there is also a useful global statement. The Hessian of $-S_i$ in physical Bloch coordinates has radial eigenvalue $1/(1 - R_i^2)$ and tangential eigenvalue $\text{arctanh}(R_i)/R_i$; both are at least one. Thus the entropy contributes at least TI to the physical-coordinate Hessian. If $T > \kappa$, Eq. (13) is strictly convex over the product of open Bloch disks, even in the presence of the linear transverse driver. The local condition $q > \kappa$ may hold at lower T and is weaker than this sufficient global-convexity bound. These results clarify why an initial high-temperature phase can be easy to optimize and why a later cooling phase can acquire many competing basins.

D. What the discrete-neighbour continuation changes

The current smooth K2000, Wishart, and Chook runs use the exact gradient of Eq. (13). A later G-set search additionally studies a surrogate direction. At update t , define

$$\widetilde{\mathbf{m}}_z^{(t)} = (1 - \lambda_t)\mathbf{m}_z^{(t)} + \lambda_t \text{sign}(\mathbf{m}_z^{(t)}), \quad 0 \leq \lambda_t \leq 1. \quad (30)$$

The term $K\mathbf{m}_z$ in the longitudinal field of Eq. (23) is replaced by $K\widetilde{\mathbf{m}}_z^{(t)}$; the local Bloch Jacobian, transverse term, and entropy derivative remain continuous. At $\lambda_t = 0$ this is the exact smooth gradient. A gradual schedule or a hard switch can increase λ_t during the run. When $\lambda_t > 0$, the resulting vector is generally *not* the gradient of the original free energy, so monotone descent of Eq. (13) is not promised. Final binary assignments are nevertheless scored with the original couplings. The G-set results that use this update must therefore be described as a distinct search variant. The frozen K2000 experiment has not applied it.

IV. NUMERICAL ALGORITHM AND WHAT A TRAJECTORY MEANS

A. Coupling scale and the implemented objective

All evaluated instances have zero local fields. Let K^{raw} be the symmetric, zero-diagonal Ising coupling matrix. For a graph with edges listed once, the implementation defines

$$\begin{aligned} c_K &= \sqrt{\frac{1}{N} \sum_{i,j} (K_{ij}^{\text{raw}})^2} \\ &= \sqrt{\frac{2}{N} \sum_{\{i,j\} \in E} w_{ij}^2}, \quad K = K^{\text{raw}}/c_K. \end{aligned} \quad (31)$$

It optimizes Eq. (13) with K , then scores binary spins against K^{raw} . The numerical values of T and Γ must therefore be read in units of the normalized interaction scale. If one multiplies the normalized free energy by c_K , the equivalent physical temperature and transverse field are $c_K T$ and $c_K \Gamma$. Coupling normalization is more than a harmless change of printed units if the temperature and driver schedules are left fixed while c_K changes.

The selected scale is 44.71018 for K2000, 0.316598 for the saved Wishart instances, and 3.162278 for the saved Chook instances. The values and their source configurations are retained with the result folders. G-set uses graph-specific normalization, and the later spectral schedule also uses the lowest eigenvalue of the normalized graph matrix.

B. Initialization, schedules, and optimization

A trial starts with B independent replicas. For replica b and site i , the two local fields are initialized as

$$a_{b,i,d}^{(0)} = h_{\text{init}} \xi_{b,i,d}, \quad \xi_{b,i,d} \sim \mathcal{N}(0, 1), \quad d \in \{x, z\}. \quad (32)$$

An isolated CPU random generator produces the stored initialization for each seed before it is transferred to the compute device. This improves replayability; it does not make distinct devices bitwise equivalent. The K2000 h_{init} is 0.1. A trial is one full population of B replicas, not one replica.

For L optimizer updates, define $\tau_t = t/(L-1)$ for $t = 0, \dots, L-1$. The simple frozen schedules have

$$T_t = T_{\min} + (T_{\max} - T_{\min})(1 - \tau_t), \quad (33)$$

$$\Gamma_t = \Gamma_{\min} + (\Gamma_{\max} - \Gamma_{\min}) \left[1 - \min\left(\frac{\tau_t}{f_\Gamma}, 1\right) \right]. \quad (34)$$

The code passes the reciprocal $\beta_t = 1/T_t$ to the free-energy kernel. For K2000 and Wishart, $f_\Gamma = 0.5$, so the transverse field vanishes halfway and the remaining updates optimize with $\Gamma = 0$. For Chook, $f_\Gamma = 1$. The G-set runs also use quadratic cooling or the spectral construction of Eq. (29), according to their graph-level plans. A schedule is applied to the objective at each update; it is not a simulated measurement history or Schrödinger evolution.

For the frozen K2000, Wishart, and Chook runs, automatic differentiation computes Eq. (23), and PyTorch RMSprop updates all field coordinates. The recorded implementation uses ordinary *coupled* weight decay, so its elementwise updates can be written as

$$\widetilde{g}_t = \nabla_{\mathbf{a}} \mathcal{F}_t + \lambda_{\text{wd}} \mathbf{a}_t, \quad v_t = \alpha v_{t-1} + (1 - \alpha) \widetilde{g}_t^{\odot 2}, \quad (35)$$

$$b_t = \mu b_{t-1} + \frac{\widetilde{g}_t}{\sqrt{v_t} + \varepsilon_{\text{opt}}}, \quad \mathbf{a}_{t+1} = \mathbf{a}_t - \eta b_t. \quad (36)$$

These equations describe the non-centered, non-Nesterov mode used in the saved plan. The operations are elementwise in the optimizer state, except for the matrix product in the gradient. Here $\varepsilon_{\text{opt}} = 10^{-8}$. Weight decay changes the numerical trajectory and thus must be reported with the learning rate and momentum. Because each replica has separate coordinates, the batch is a parallel collection of independent optimizations under the same graph and schedule. The implementation backpropagates a sum of per-replica free energies; it does not divide the gradient by B .

For clarity, the complete frozen-run logic is:

1. Load and validate the graph or planted Ising instance; form normalized couplings $K = K^{\text{raw}}/c_K$.
2. For each declared seed, construct $B \times N \times 2$ Gaussian initial fields and arrays $\{T_t, \Gamma_t\}_{t=0}^{L-1}$.

3. For every update, map fields through Eq. (15), evaluate Eq. (13), and apply Eq. (35)–(36) or the declared alternative optimizer.
4. After exactly L updates, decode all B final states and rescore each binary vector with the raw instance objective.
5. Record the best final replica for that seed; aggregate these trial endpoints using the declared mean, best, gap, and exact-hit metrics.

The K2000 population experiment uses nested prefixes of the same initialized 8,192-replica population for each seed. This controls the random population when B changes: a 128-replica run uses exactly the first 128 initial states of that seed’s 8,192-replica run, and similarly for larger sizes. It does not imply that their later trajectories remain prefixes, since floating-point batching and optimizer execution can differ.

C. Binary readout and scoring

The implemented final readout is

$$\hat{s}_{b,i} = \begin{cases} +1, & m_{z,b,i}^{(L)} > 0, \\ -1, & m_{z,b,i}^{(L)} \leq 0. \end{cases} \quad (37)$$

This is equivalent to rounding the probability $p_{b,i} = (1 + m_{z,b,i})/2$ as used by the solver, including the tie at $p = 1/2$. It yields a feasible binary vector for every replica. For MaxCut, we evaluate Eq. (6); for planted Ising instances we evaluate $E_I(\hat{\mathbf{s}}_b)$ and compare it with the supplied planted energy. No local improvement, bit flipping, coordinate splicing, or selection of an intermediate iterate is part of the frozen endpoint protocol.

This distinction is crucial for interpreting trajectory figures. The stored K2000 trace first selects the final best replica of the best endpoint trial, then replays that *same replica* from its initial field to the final update. The curve is not the maximum cut over the population at every step. Its temporary decreases are possible and do not contradict final best-of-population selection. The entropy trace is likewise the sum of site entropies for that coherent replica, divided by N when plotted.

D. Arithmetic and storage

For a dense graph, one longitudinal field evaluation is a batched matrix product $\mathbf{m}_z K$, costing $O(BN^2)$ arithmetic, with $O(N^2 + BN)$ persistent storage before optimizer buffers and temporary autograd tensors. An edge-list or sparse matrix contraction can reduce arithmetic to $O(B|E| + BN)$ and graph storage to $O(|E|)$, subject to device support and implementation details. K2000 is complete, so $|E| = N(N - 1)/2$ and the dense formulation is natural. Large replica counts improve the number

of basins explored but increase memory and arithmetic nearly linearly in B . Hence solution quality at $B = 8,192$ cannot be read as the quality of a single cheap trajectory, and unmatched wall-clock measurements across methods are not evidence of a speedup.

V. EXPERIMENTAL MATERIAL AND EVALUATION PROTOCOL

A. Instance families and reference values

The saved evidence covers two MaxCut families and two planted Ising families. G-set G1–G54 consists of 54 MaxCut graphs with varied size and topology; the supplied target file serves as the versioned cut reference [10]. K2000 is a complete signed 2,000-vertex graph with $2,000 \times 1,999/2 = 1,999,000$ undirected edges [11]. Its stored target cut is 33,337. A stored target is a known-quality comparison value, not in itself a certificate of global optimality.

B. Planted data and their construction

The Wishart bank contains one dense 500-spin instance at each $\alpha = M/N \in \{0.1, 0.2, \dots, 1.0\}$, with $M = 50, 100, \dots, 500$. Its generator centers M Gaussian columns to make them orthogonal to a planted spin direction, forms a zero-diagonal coupling matrix from their Gram matrix, and applies a random spin gauge [12]. The Chook bank instead contains one sparse, periodic 32×32 lattice at each requested $p(C_3) \in \{0.0, 0.1, \dots, 1.0\}$. Edge-disjoint local tiles are drawn from the C_2/C_3 mixture so that a common planted spin assignment minimizes their sum; a gauge hides that assignment [13]. Each saved file carries its planted spins and energy. These are fixed local realizations of the published families, not the unpublished matrices used in the cited figures; each plotted parameter point describes one file rather than an ensemble average.

C. Scoring and endpoint interpretation

For a MaxCut candidate, the recorded residual is

$$\Delta_C = C_{\text{ref}} - C(\hat{\mathbf{s}}). \quad (38)$$

If $\Delta_C = 0$, the stored reference was attained. If a new cut exceeded it, the residual would be negative; none of the retained G-set results exceeded its supplied references. For planted instances, the recorded energy excess and relative error are

$$\Delta_E = E(\hat{\mathbf{s}}) - E_0, \quad \varepsilon = \frac{|E(\hat{\mathbf{s}}) - E_0|}{|E_0|}. \quad (39)$$

An exact hit means the final rescored energy equals the supplied planted energy within the stored benchmark’s

exact scoring convention. A hit *trial* means at least one replica in that trial hits. The number of such trials must not be confused with the number of hitting replicas; the available compact paper records preserve the winning endpoint and trial hit status, not every candidate population for every benchmark.

D. Frozen QeFEM and LQA comparison

The K2000, Wishart, and Chook experiments use selected, frozen family-level configurations from the portable `qefem_prof` recheck plans. They were chosen after exploratory tuning and therefore should be interpreted as evaluations of tuned procedures on the stored instances, not as a pristine held-out generalization study. Every QeFEM trial evolves 8,192 replicas for the canonical comparison. The K2000 and Wishart experiments use five seeded trials per instance; Chook combines two predeclared five-seed batches for ten trials per instance. The CPU, float32 portable evaluation is recorded with source revision `b14f73c53305b55242b418ef2013fba0ac3cce54` and input hashes.

The comparator is an independently implemented LQA run based on the classical product-state gradient method of Bowles *et al.* [6]. Its local states are pure and parameterized by one angular variable per spin; one trial evolves one such state. The saved runs use Adam with $(\beta_1, \beta_2) = (0.9, 0.999)$, zero weight decay, and five or ten trials matched in count and seed labels to the corresponding QeFEM trials. K2000 uses 5,000 updates, learning rate 1, and driver parameter $g = 0.1$; Wishart and Chook use 500 updates, learning rate 2, and $g = 1$. The initial range parameter is $f = 0.1$. The planted-instance value $g = 1$ is a local benchmark choice, not a value specified in the published LQA figure caption. Thus these results compare the declared saved implementations; they should not be presented as a reproduction of every published LQA setting. Equal trial counts are also not equal state counts or equal arithmetic work.

E. G-set: adaptive attainment and fresh-seed confirmation

The G-set evidence has a different design. An earlier smooth-gradient study retained reference cuts on 23 of the 54 graphs. A later search revisited only the 31 unmatched instances. Its controller proposed seven initial configurations per graph, including transferred spectral schedules, exact manual gradients, and hard or gradual discrete-neighbour updates. Up to six regional proposals refined a graph when discovery had not yet hit its reference. Candidate selection used discovery and promotion data; the chosen configuration was frozen before an evaluation on new seed bases. The full search record reports 990 trial executions and 502,016 replica candidates across discovery, promotion, and frozen evaluation, excluding

historical effort. Table III specifies the stage budgets.

The frozen evaluation budget depends on each graph’s earlier gap. Graphs within two cut units receive 4,096 replicas and three new seeds; others receive 2,048 replicas and two. The older 23 hits remain prior attainments rather than being rerun as part of this 31-graph search. Consequently the cumulative 33/54 figure describes what has been *attained across all saved searches*. It is not an estimate that a single frozen universal configuration succeeds on 33 of 54 graphs or that a random seed has success probability 33/54.

F. Reproducibility and the scope of comparisons

Every frozen endpoint is tied to an instance hash, configuration, seed, numerical precision, and device record. The saved paper workspace stores endpoint CSVs, selected coherent trajectories, reports, and configuration JSON files; full raw candidate populations from earlier runs were not copied into it. The G-set search audit independently rescored saved populations in its own source workspace. Results shown in the main planted comparison use the same fixed instance files for both methods.

We report cut quality, energy error, and exact-hit counts. We do not calculate time-to-solution, speedup, or energy efficiency from the mixed resource protocols: QeFEM uses thousands of simultaneous states, LQA uses one, the G-set search is adaptive, and archived FEM/dSB notebook output came from a separate run environment. Within those limits, the stored data still support specific, useful conclusions about attainable quality and where the current method fails.

VI. RESULTS AND THEIR INTERPRETATION

A. A compact view of the frozen comparison

Table IV gives the aggregate result before the instance-level analysis. A mean over an instance family gives each of its fixed files equal weight. The planted exact-hit count is the number of trial populations whose best final candidate reaches the planted energy; it is not a count of all hitting replicas.

The aggregate advantage over this LQA implementation is consistent across the planted files, but the scale of that advantage depends on the family and its parameter. The detailed plots and tables below show where the exact hits arise. K2000 reveals a different pattern: a measurable advantage over the saved LQA endpoints, coupled to a persistent 44-unit shortfall against the stored cut reference.

TABLE I. Frozen QeFEM schedules and budgets. A trial includes all replicas. The driver reaches zero at fraction f_T of the run. Temperature follows a linear decrease in T , not a linear increase in $\beta = 1/T$.

Family	updates	replicas	trials	h_{init}	$T_{\text{max}} \rightarrow T_{\text{min}}$	$\Gamma_{\text{max}} \rightarrow 0$	f_T
K2000	1,000	8,192	5	0.10	$1 \rightarrow 10^{-4}$	$1 \rightarrow 0$	0.50
Wishart	500	8,192	5	0.10	$1 \rightarrow 10^{-4}$	$0.3 \rightarrow 0$	0.50
Chook	500	8,192	10	0.05	$2 \rightarrow 10^{-3}$	$1 \rightarrow 0$	1.00

TABLE II. Frozen QeFEM RMSprop parameters. All three use non-centered RMSprop, $\varepsilon_{\text{opt}} = 10^{-8}$, and float32 fields; λ_{wd} is coupled weight decay. The RMS scale is defined in Eq. (31).

Family	learning rate η	α	momentum μ	λ_{wd}	RMS scale c_K
K2000	0.03	0.98	0.91	0.010	44.71018
Wishart	0.03	0.56	0.63	0.013	0.316598
Chook	0.03	0.90	0.90	0.001	3.162278

B. G-set: broad attainment with a measurable later gain

The smooth-gradient study reached 23 of 54 supplied G-set reference cuts. The later search on the remaining 31 produced ten additional matches, for 33 cumulative matches. Eight of the ten new matches were recovered after freezing a selected configuration and changing to the final evaluation seeds: G15, G16, G18, G19, G21, G22, G24, and G25. G17 and G28 were attained during search or promotion but not on their fresh evaluation seeds. Among the 31 revisited graphs, the retained gap decreased on 27.

The improvements extend beyond the ten new matches. Most revisited graphs move closer to their references even without reaching them. The remaining failures are also concentrated: G14, G29, G30, G31, and G51 are one cut unit below their stored targets; G23, G26, and G40 are two units below. The largest residuals are on G35–G38, with gaps 13, 16, 18, and 14 respectively. Appendix C lists every G1–G54 target, attained cut, gap, and evidence status. The 33 matches do not establish an equal-budget advantage over another solver because the cumulative search incorporated multiple graph-specific proposals.

C. K2000: a strong cut, a population plateau, and an open gap

K2000 tests a dense signed objective with nearly two million couplings. The five frozen 8,192-replica QeFEM trials yield cuts 33,274, 33,274, 33,274, 33,293, and 33,291. Their mean is 33,281.2 and sample standard deviation is 9.88 cut units. The best cut is 44 below the stored 33,337 reference, a residual of 0.132% relative to that reference. The corresponding five LQA endpoints all equal 33,249, giving gap 88. The smaller QeFEM gap is an attainable-quality result under these frozen method settings; it is not a matched-time win because one QeFEM trial evolves

8,192 states and one LQA trial evolves one.

The nested population study diagnoses how much of the gain comes from additional initial conditions. At 128 replicas, the mean cut over five seeds is 33,252 and the best is 33,255. At 2,048 replicas, these rise to 33,272.6 and 33,293. Increasing the population further to 8,192 raises the mean to 33,281.2 but leaves the best at 33,293. Thus more replicas improved average trial quality within the explored range, while four times as many replicas after 2,048 did not close the final best-cut gap. Figure 2 makes both facts visible.

The coherent final-winner trajectory in Fig. 3 provides an additional view of the search. Its discrete reference gap falls by orders of magnitude over the run, while local entropy per spin decreases overall. The early entropy dip and rebound show that neither entropy nor discrete cut changes monotonically under RMSprop and a changing objective. The driver reaches zero halfway; the final portion continues to refine the same candidate under a classical objective. This selected trajectory illustrates a mechanism visible in one successful trial; it does not describe the distribution of all 8,192 replicas.

Taken together, the K2000 results support a specific and bounded claim. The frozen mixed-state solver produces a high cut and improves on the saved one-state LQA baseline, but the explored population scaling does not reach the reference. A new schedule or the G-set discrete update might change that outcome; neither has been evaluated here and neither can be credited for the frozen K2000 result.

D. Wishart: lower error on every fixed planted instance

The Wishart mean relative energy error is 4.0717% for QeFEM and 6.9411% for the saved LQA comparator. The difference is 2.8693 percentage points, or a 41.3% reduction relative to the latter mean. QeFEM has lower mean error at each of the ten requested α values. Exact

TABLE III. Later G-set search budgets, copied from the completed search report. A “candidate” is one replica endpoint, not an independent graph or a universally tuned hyperparameter.

Stage	trials	replica endpoints	per-setting budget
Discovery	786	100,608	128×2 seeds
Promotion	124	126,976	$1,024 \times 2$ seeds
Frozen evaluation	80	274,432	$2,048 \times 2$ or $4,096 \times 3$
Total	990	502,016	later search only

TABLE IV. Aggregate frozen endpoints. “Best” means largest final cut on K2000 or smallest relative energy error on a planted instance. The methods use their declared, unequal internal state budgets.

Family	method	trials	mean	best; exact hits
K2000	QeFEM	5	cut 33,281.2	cut 33,293; gap 44
K2000	LQA	5	cut 33,249.0	cut 33,249; gap 88
Wishart	QeFEM	50	$\varepsilon = 4.0717\%$	0%; 7/50
Wishart	LQA	50	$\varepsilon = 6.9411\%$	2.0199%; 0/50
Chook	QeFEM	110	$\varepsilon = 0.2186\%$	0%; 60/110
Chook	LQA	110	$\varepsilon = 1.1086\%$	0%; 10/110

planted-energy hits occur in two of five trials at $\alpha = 0.9$ and all five at $\alpha = 1.0$; the saved LQA runs have none across the 50 trials. The late improvement of QeFEM after a larger error at $\alpha = 0.8$ is consistent with an easy-hard-easy pattern in this particular ten-file series, but these fixed files alone do not estimate an ensemble-wide hardness curve [12].

E. Chook: a wide planted-hit region

Across the eleven tile-planted files, QeFEM has equal-instance mean relative error 0.2186%; the saved LQA mean is 1.1086%. The absolute difference is 0.8900 percentage points, corresponding to an 80.3% relative reduction. QeFEM has lower mean error on every file. It reaches the planted energy in all ten trials for every requested $p(C_3)$ from 0.5 through 1.0, accounting for 60 of 110 trial hits. LQA has four hits at 0.9 and six at 1.0, for ten of 110 overall. The requested $p(C_3)$ is a generator setting; the realized tile fraction of a finite instance can differ slightly.

The two measures capture complementary outcomes. A lower mean error on every file demonstrates consistent improvement over the declared comparator, while exact hits distinguish a merely close endpoint from a solved planted instance. The same plot also shows where the gap has not closed: Wishart $\alpha \leq 0.8$ yields no exact hit in these five-seed batches.

VII. DISCUSSION

A. What the experiments establish

The evidence identifies a useful algorithmic property: the same factorized mixed-state construction produces competitive binary assignments on sparse and dense Max-Cut graphs and on two structurally different planted Ising families. On the saved planted files, the QeFEM configuration has lower error than the declared LQA configuration in every instance, and it returns exact planted energies in seven Wishart and 60 Chook trial populations. The G-set continuation adds ten previously unattained reference cuts, eight of them recovered with frozen settings on fresh seeds. These are concrete strengths, not claims of quantum hardware advantage or universal optimality.

The dynamics help explain why those strengths are plausible. The product-state expectation is cheap enough for a large replica batch; high-temperature entropy gives a smooth starting regime; the driver offers an additional local direction during part of the search; and cooling lets the longitudinal coupling landscape determine the final binary choices. Spectral scaling provides a principled coordinate for the onset of longitudinal instability on zero-field graphs. None of these mechanisms guarantees the best basin, but each gives a testable design variable rather than an opaque postprocessing step.

The K2000 result imposes an equally clear boundary. A best cut of 33,293 is strong relative to the saved 33,249 LQA cut, yet it remains 44 below the stored 33,337 reference. The best cut stops improving in the recorded nested population sweep after 2,048 replicas. A larger population alone therefore has no demonstrated route to reference attainment in these data. The available FEM/dSB notebook outputs reach 33,337 under their own protocols.

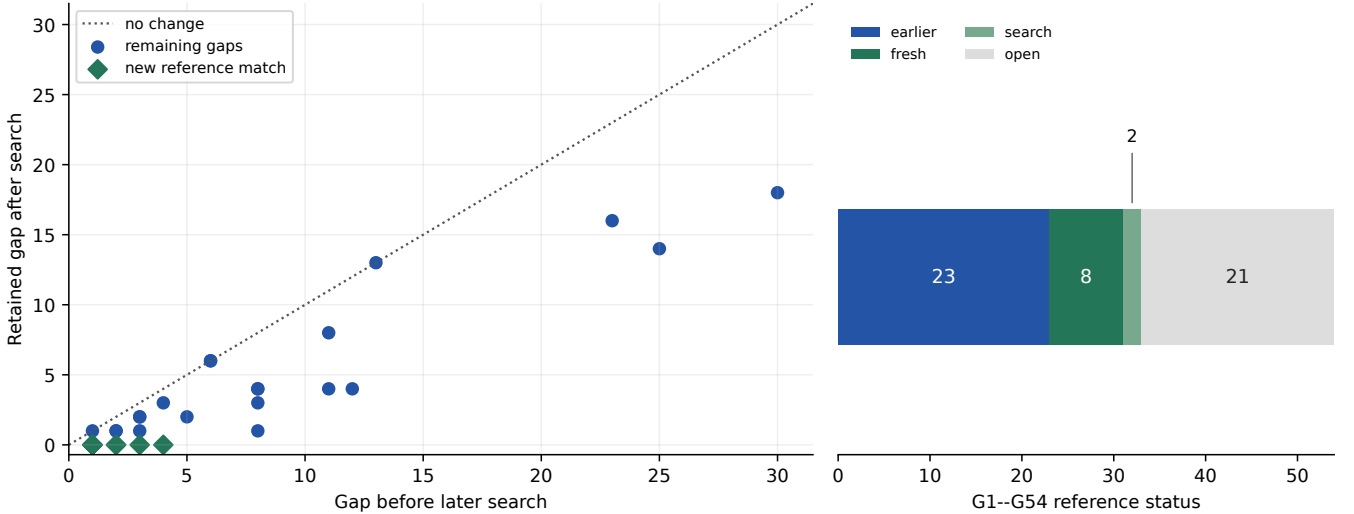


FIG. 1. **Targeted QeFEM updates extend attainment on previously unresolved graphs.** Each point in the left panel is one of the 31 graphs below its supplied reference before the later search. Points below the diagonal improved; diamonds on the horizontal axis are the ten new matches. The right panel separates the accumulated 54-graph inventory into 23 prior matches, eight new matches recovered with fresh seeds, two found only during search, and 21 unmatched graphs. Search budgets were adaptive, so the stacked bar is an attainment record, not a universal success rate.

TABLE V. All five frozen K2000 trials. The paired rows share a seed label; the algorithms use different state representations and random-input shapes. Both methods are rescored on the same raw graph.

trial	seed	QeFEM cut	gap	LQA cut	gap
1	3371835386	33274	63	33249	88
2	3700479728	33274	63	33249	88
3	1934420243	33274	63	33249	88
4	716463040	33293	44	33249	88
5	424211264	33291	46	33249	88

The correct conclusion is that the present QeFEM K2000 configuration does not match those attained cuts; the evidence does not establish that the state family is incapable of matching them under a different schedule or update. Conversely, theoretical inclusion of a classical endpoint does not prove a finite QeFEM run must outperform tuned FEM dynamics. The endpoint proposition in Sec. III makes that distinction exact.

B. Why this can matter outside the present benchmarks

Many scheduling, allocation, routing, and design tasks admit QUBO or Ising formulations [3]. Once such a formulation is available, the variational state, entropy evaluation, readout, and batch optimizer in QeFEM do not depend on the application’s name; the objective enters through its fields and pair couplings. That separation makes the method attractive for an optimization platform that must support many binary models on accelerator

hardware. The experiments here support the *technical possibility* of scaling a single implementation from sparse G-set graphs to a 2,000-spin dense graph and to planted suites with known targets. They do not yet measure business value, deployment reliability, constraint satisfaction after QUBO penalty tuning, or total energy and monetary cost on an industrial workload. Those questions require application-specific experiments.

The distinction also shapes how performance should be communicated. A result such as 60/110 planted-energy hits conveys something different from an average percent error; together they show how often the method fully solves a fixed file and how far the remaining runs miss. The G-set count is compelling as an attainment inventory, but its adaptive search budget must accompany it. On K2000, a 0.132% reference-relative residual sounds small, but the absolute 44-unit gap is the more direct optimization fact on a signed-weight graph. Clear plots and complete captions can present a strong algorithm without converting unlike protocols into an unsupported ranking.

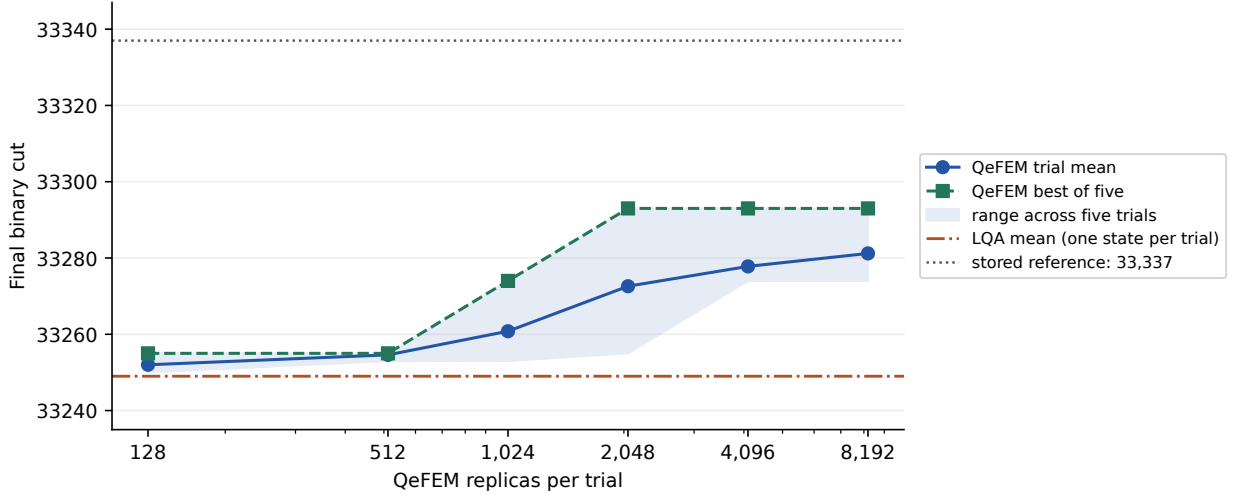


FIG. 2. **Population search improves K2000 cut quality, with diminishing best-cut returns.** QeFEM’s mean, range, and best final cut across five paired seeds are shown at each nested population size. The saved LQA mean (one state per trial) and the 33,337 stored target are horizontal references. The best QeFEM cut rises from 33,255 at 128 replicas to 33,293 by 2,048, then remains there through 8,192. No point in this figure implies a matched-compute comparison.

C. Limitations and decisive next experiments

The product ansatz omits intersite correlations and entanglement. Low-temperature spin-glass landscapes can have many metastable basins, and the optimizer follows a finite, schedule-dependent path rather than the instantaneous variational minimum. The manual discrete-neighbour update can further improve attainable cuts, but it is a surrogate vector field, not the exact free-energy gradient. No general convergence or approximation-ratio theorem is claimed for the complete practical solver. In particular, an observed local spectral instability does not imply global MaxCut optimality.

Several limitations are experimental. The plotted G-set total accumulates adaptive tuning, historical runs, and fresh-seed confirmation. K2000 has one graph and five canonical QeFEM seeds; the plotted planted series has one fixed file at each parameter. Family-level hyperparameters were selected after exploration of related data. The saved LQA comparison evolves one state per trial, whereas QeFEM evolves 8,192, and its planted settings include a local driver choice. No wall-clock-normalized or candidate-normalized ranking is possible from these endpoints alone. Even the FEM/dSB K2000 notebook is a contextual source rather than a matched three-way protocol; its saved output predates a later device-selection code change, so the exact numbers should be freshly re-produced before they become a formal new head-to-head figure.

A decisive K2000 study would freeze graph input, precision, device, scoring code, and a predeclared total-work budget across QeFEM, FEM, and dSB. It would vary the QeFEM schedule and the G-set-derived manual-discrete update under separate, recorded hyperparameter searches;

then it would evaluate selected settings on fresh seeds without moving the goalposts. Reporting per-replica cut distributions, hit counts, time-to-target, and memory would determine whether the remaining gap is a schedule issue, a basin-coverage issue, or a more structural limitation. The current paper does not include such a trial, and it does not extrapolate the G-set gains to K2000.

VIII. CONCLUSION

QeFEM gives a mathematically explicit route from a binary Ising objective to a differentiable, annealed search over local mixed states. The quantum relative entropy supplies the free-energy principle; the product ansatz makes it computationally tractable; the Bloch-field map guarantees valid local states; and binary readout returns ordinary feasible solutions. The theory locates a graph-dependent longitudinal instability scale and shows exactly where the later discrete-neighbour search departs from the smooth gradient.

Under the saved protocols, the method reaches 33 of 54 G-set reference cuts across accumulated searches, improves the declared LQA comparison on every fixed Wishart and Chook instance, and gives a 33,293 K2000 cut while leaving a 44-unit reference gap. These results make a coherent case for QeFEM as a flexible classical optimization framework with strong solution quality on several large binary landscapes. They also identify the next falsifiable question: whether the new search dynamics and carefully matched resources can close the dense K2000 gap without relying on ever larger replica populations.

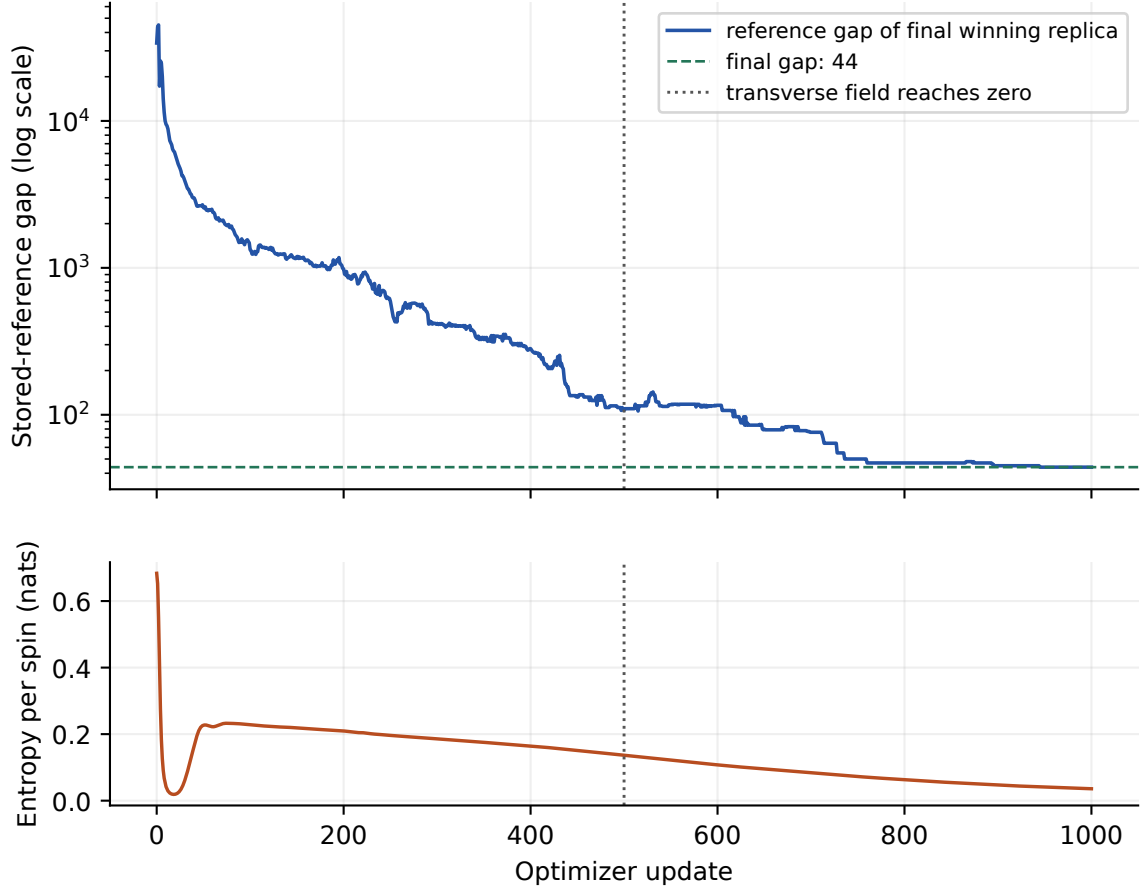


FIG. 3. **Annealed mixed-state optimization can turn a poor initialization into a near-reference binary solution.** The upper panel shows the stored-reference gap on a logarithmic scale for one coherent replica: the final winner of the best 8,192-replica K2000 trial. The lower panel shows its mean single-site entropy. The dotted vertical line marks the end of the transverse-field schedule; the dashed horizontal line marks the final gap of 44. This is one replica’s replay, not a per-step maximum over the population.

AUTHOR CONTRIBUTIONS

R.D. conceived and developed the core QeFEM method, implemented the original solver, designed the analysis workflow, and drafted the manuscript. F.P. contributed the nonlinear reparameterization, optimizer and schedule refinement, and benchmark testing. Both authors interpreted the results and take responsibility for the final text.

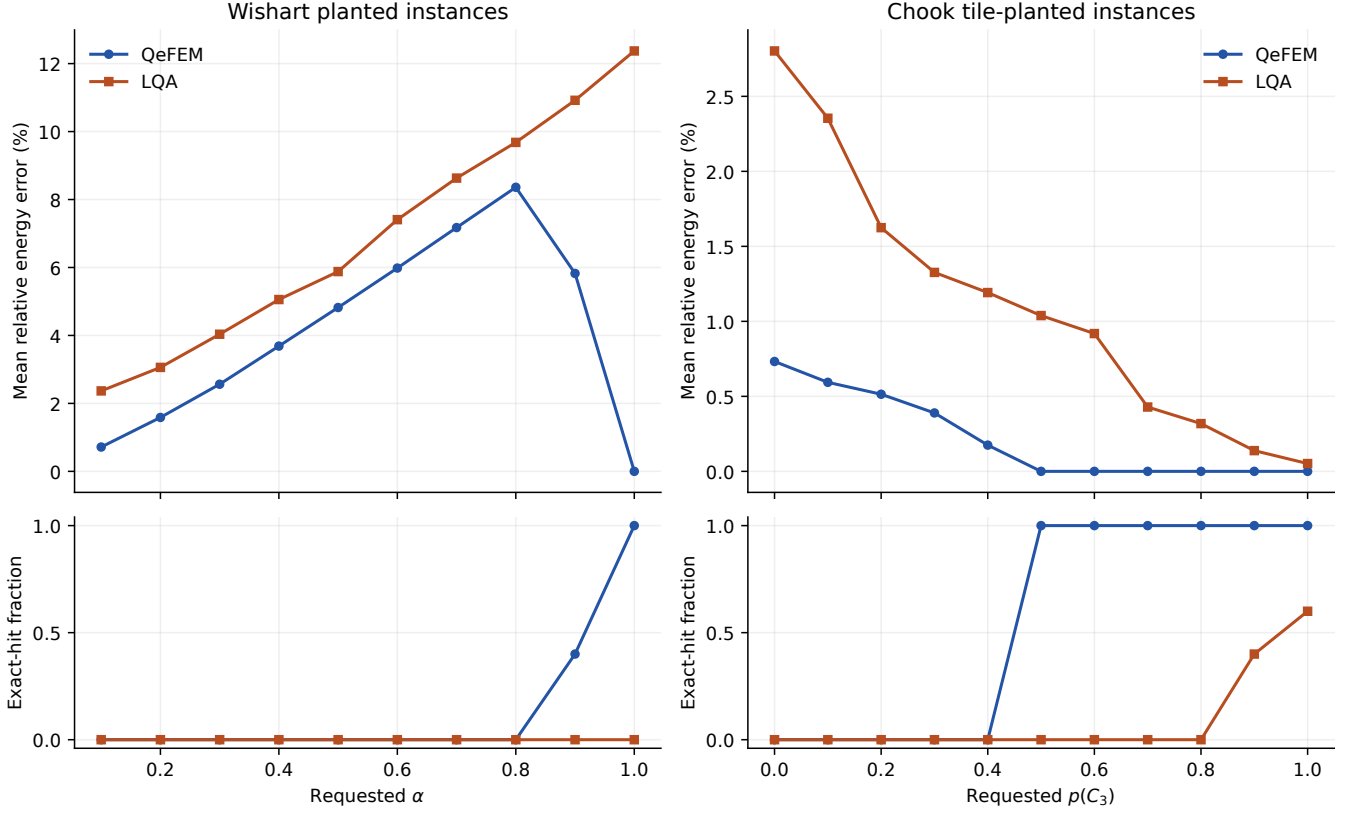


FIG. 4. Under the saved configurations, QeFEM improves quality throughout both planted suites and reaches exact solutions more often. Top panels show trial-mean relative energy error on each fixed Wishart and Chook instance. Bottom panels show the fraction of trial populations whose best final replica has exactly the planted energy. There are five trials per Wishart instance and ten per Chook instance. The curves connect different fixed files and are visual guides, not uncertainty bands or estimates over newly sampled instances.

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Appendix A: Detailed derivations and limiting cases

1. From quantum relative entropy to a product free energy

The Gibbs target of Eq. (9) is full rank for $T > 0$. Starting from the definition of relative entropy,

$$\begin{aligned} D_{\text{KL}}(\rho \parallel \sigma_{T,\Gamma}) &= \text{Tr}(\rho \log \rho) - \text{Tr}(\rho \log \sigma_{T,\Gamma}) \\ &= -S(\rho) + \frac{\text{Tr}(\rho H_\Gamma)}{T} + \log Z_{T,\Gamma}. \end{aligned} \quad (\text{A1})$$

Multiplication by T immediately gives Eq. (10). The omitted constant in an optimization is $-T \log Z$, which depends on (T, Γ) but not on the variational fields. Omitting it from gradient calculations is valid even though its value changes along the schedule. The exact Gibbs minimum is outside the factorized family in general, so the restricted minimum only bounds the exact free energy from above.

For a product state $\rho = \bigotimes_i \rho_i$, the logarithm separates as $\log \rho = \sum_i I \otimes \cdots \otimes \log \rho_i \otimes \cdots \otimes I$, giving $S(\rho) = \sum_i S(\rho_i)$. For $i \neq j$,

$$\text{Tr}[\rho(Z_i Z_j)] = \text{Tr}(\rho_i Z_i) \text{Tr}(\rho_j Z_j) = m_{z,i} m_{z,j}. \quad (\text{A2})$$

The transverse expectation is $\text{Tr}(\rho X_i) = m_{x,i}$. These identities give Eq. (13) without sampling any spin configuration.

2. The local state, entropy, and field map

Let $A_i = u_i X_i + v_i Z_i$. The Pauli anticommutation relation gives $A_i^2 = (u_i^2 + v_i^2)I = r_i^2 I$. Separating even and odd terms of the exponential series,

$$e^{A_i} = \cosh r_i I + \frac{\sinh r_i}{r_i} A_i. \quad (\text{A3})$$

Dividing by $\text{Tr} e^{A_i} = 2 \cosh r_i$ produces $\rho_i = \frac{1}{2}[I + (\tanh r_i / r_i) A_i]$. Thus the Bloch radius is $R_i = \tanh r_i$, and the eigenvalues are $(1 \pm R_i)/2$. Substituting those eigenvalues into $-\text{Tr} \rho_i \log \rho_i$ yields Eq. (12). Alternatively, from $\log \rho_i = A_i - \log(2 \cosh r_i)I$,

$$\begin{aligned} S_i &= -\text{Tr}(\rho_i A_i) + \log(2 \cosh r_i) \\ &= \log(2 \cosh r_i) - r_i \tanh r_i, \end{aligned} \quad (\text{A4})$$

because $\text{Tr}(\rho_i A_i) = r_i R_i$.

The apparent singularity at $r = 0$ in $g(r) = \tanh r / r$ is removable. Expanding $\tanh r$ around zero gives

$$g(r) = 1 - \frac{r^2}{3} + \frac{2r^4}{15} + O(r^6), \quad \text{sech}^2 r = 1 - r^2 + \frac{2r^4}{3} + O(r^6). \quad (\text{A5})$$

These limits justify a smooth origin and the numerical safe branch in the implementation.

3. Gradient and Hessian in physical and field coordinates

Writing $\mathbf{m} = (m_x, m_z)$ and $R = \|\mathbf{m}\|$, the scalar derivative of Eq. (12) is

$$\frac{dS}{dR} = -\frac{1}{2} \log \frac{1+R}{1-R} = -\operatorname{arctanh} R. \quad (\text{A6})$$

As $\nabla_{\mathbf{m}} R = \mathbf{m}/R$ away from the origin, $\nabla_{\mathbf{m}}(-TS) = T \operatorname{arctanh} R \mathbf{m}/R$. Adding the problem and driver gradients gives Eqs. (20)–(21). To move to unconstrained fields, differentiate $\mathbf{m} = g(r)\mathbf{a}$:

$$d\mathbf{m} = g(r)d\mathbf{a} + g'(r)\mathbf{a} \frac{\mathbf{a}^\top d\mathbf{a}}{r}. \quad (\text{A7})$$

This is Eq. (18). In the radial direction $\mathbf{a}/r$, the Jacobian eigenvalue is $g(r) + rg'(r) = d(\tanh r)/dr = \operatorname{sech}^2 r$; in every tangential direction it is $g(r)$. Since $J\mathbf{a} = \operatorname{sech}^2 r \mathbf{a}$, the chain rule applied to Eq. (22) gives Eq. (23). No derivative of a sign function appears in this exact smooth calculation.

For completeness, the physical-coordinate Hessian of the negative single-site entropy has the radial and tangential eigenvalues

$$\lambda_{\text{rad}} = \frac{1}{1-R^2}, \quad \lambda_{\text{tan}} = \frac{\operatorname{arctanh} R}{R}. \quad (\text{A8})$$

They follow by differentiating the radial vector field $\nabla(-S) = (\operatorname{arctanh} R/R)\mathbf{m}$. Since $\operatorname{arctanh} R \geq R$ for $0 \leq R < 1$, both are at least one. The graph contribution to the Hessian acts only on longitudinal coordinates and equals K . Thus $T > -\lambda_{\min}(K)$ is a sufficient condition for strict convexity of the physical-coordinate objective. This does not imply convexity in the nonlinear field coordinates $\mathbf{a}$; reparameterization may change curvature even while preserving finite-field stationary points.

4. Longitudinal instability and spectral schedule inversion

When $h = 0$ and $\mathbf{m}_z = 0$, the stationary local x component is $m_x = \tanh(\Gamma/T)$. The graph term has quadratic expansion $\frac{1}{2}\mathbf{m}_z^\top K \mathbf{m}_z$. Holding the stationary transverse component fixed for a longitudinal perturbation, the local entropy curvature contributes

$$T \frac{\operatorname{arctanh} m_x}{m_x} = \frac{\Gamma}{\tanh(\Gamma/T)} = q(T, \Gamma). \quad (\text{A9})$$

The mixed x – z Hessian vanishes at $m_z = 0$ by symmetry, so Eq. (28) is also the relevant longitudinal block of the full Hessian. If the lowest eigenvalue of K is $-\kappa$, the quadratic form first loses positive definiteness at $q = \kappa$. Setting $\rho = \Gamma/T$ gives $q = T\rho/\tanh \rho$. Algebraic inversion yields $T = q \tanh \rho/\rho$ and $\Gamma = q \tanh \rho$, as used in the graph-specific G-set schedule. The stability calculation assumes zero local fields and concerns a local stationary solution; it is not a global approximation guarantee.

5. Quadratization example and its penalty requirement

The formal reach of a QUBO solver extends to higher-order binary terms when auxiliary variables are introduced, although that extension is not benchmarked here. For $x, y, z \in \{0, 1\}$, the penalty

$$P(xy - 2xz - 2yz + 3z), \quad P > 0, \quad (\text{A10})$$

is zero when $z = xy$ and strictly positive for the other assignments. For example, $(x, y) = (0, 0)$ gives penalty $3Pz$; $(1, 0)$ or $(0, 1)$ gives Pz ; and $(1, 1)$ gives $P(1 - z)$. To preserve the optimum of a surrounding objective, P must exceed the maximum possible improvement gained by violating the intended relation. Repeated substitutions can increase variable count and coefficient spread. Consequently QUBO encodability alone is not a performance claim for a new application.

Appendix B: Protocol, source, and comparator details

1. Exact saved-plan provenance

The canonical K2000, Wishart, and Chook configurations are read from the following frozen plan files in the companion `qefem_prof` repository:

- `benchmarks/k2000/results/portable_accuracy_recheck/final_k2000/plan.json`;
- `benchmarks/wishart/results/portable_accuracy_recheck/final_wishart/plan.json`;
- `benchmarks/chunk/results/portable_accuracy_recheck/final_chook_a/plan.json` and `final_chook_b/plan.json`.

The paper workspace retains the parsed configurations and endpoint rows under `Paper/results/`. It records the Python, NumPy, and PyTorch versions, the numerical device, dtype, input hashes, and the declared source revision in `results/experiment_environment.json` and the case-level configuration and manifest files. The frozen K2000 QeFEM job was selected from `scale_k2000/momentum_fast`. That selection predates the later G-set manual-discrete search. No G-set spectral or discrete-neighbour setting was silently transferred to the plotted K2000 results.

Each K2000 seed is obtained from the declared base seed 50,000 by the repository’s deterministic derivation function. The resulting five seed values are listed in Table V; the LQA rows share those labels. Wishart and Chook likewise retain their exact seed arrays in each instance’s `configuration.json`. A seed controls initial pseudo-random fields, not the quality of the optimum itself. Different algorithms map the same integer to different state shapes, so a shared label is useful for auditing but not a proof of paired random trajectories.

2. The G-set search variants

The smooth G-set stage used a product-state objective with exact automatic differentiation. The later remaining-gap search used both an exact manual evaluation of Eq. (23) and the surrogate direction of Eq. (30). Its spectral schedule computes κ from the smallest eigenvalue of the normalized graph coupling matrix, constructs a decreasing q/κ curve, and chooses a decreasing ratio $\rho = \Gamma/T$. Equation (29) then generates the temperature and driver arrays. Most transferred discrete-update proposals used SGD with momentum 0.95 and weight decay 0.01; exact proposals retained their previous optimizer. This is a graph-specific search, not one fixed set of hyperparameters across all G-set graphs.

For the discrete mode, λ_t can increase linearly between declared start and end fractions or switch abruptly when those fractions are equal. The implementation changes only the neighbor magnetizations inside $K\widetilde{\mathbf{m}}_z$ and leaves the local Bloch Jacobian and entropy contribution continuous. The final configuration is always decoded and rescored against the original graph. The full controller, per-graph plan, proposal registry, and rescore audit are saved under `benchmarks/Gset/results/remaining_discrete/`; its completed analysis is the source of the 54-instance table in Appendix C.

3. How the FEM and dSB context was handled

The accompanying FEM repository contains a K2000 notebook with saved cell output for separate FEM and dSB sweeps. The current notebook source sets 1,000 replicas for each method and nominal update budgets from 100 through 100,000. Its saved output reports the best cuts in Table VI. These values are included so the reader can see that the current QeFEM gap is a real competitive limitation, not because the notebook is a controlled three-way trial. The notebook itself states that its saved outputs predate a later device-selection code update; rerunning is needed to align saved values with the current MPS/CPU implementation. The published FEM paper reports its own K2000 protocol and figure [5]; its figure should not be numerically identified with this local notebook’s later output.

The FEM notebook uses RMSprop-like momentum settings $\eta = 0.03$, $\alpha = 0.56$, momentum 0.63, weight decay 0.013, $T_{\max} = 1.16$, $T_{\min} = 6 \times 10^{-5}$, and a graph-dependent normalization in its classical FEM implementation. Its dSB implementation uses step size $\Delta t = 1.25$ and a sign-dependent coupling in a position–momentum update. These are genuine algorithmic and protocol differences. Figure 2 consequently compares only the frozen QeFEM population study with its explicitly declared saved LQA baseline; it does not mix a FEM/dSB sweep into one curve.

4. Figure data and rebuild command

All four figures in this manuscript are generated by `detailed/make_figures.py`. The script performs read-only loads of saved endpoint, trajectory, and G-set analysis files; it executes no optimizer. It writes vector PDF and

TABLE VI. Selected best cuts printed by the archived FEM K2000 notebook, with 1,000 FEM or dSB trajectories at each independent update budget. These are contextual outputs, not matched comparisons with the separate 8,192-replica, five-trial QeFEM CPU evaluation. The stored reference is 33,337.

updates	archived FEM best	archived dSB best
100	33,133	32,993
200	33,214	33,155
400	33,275	33,248
600	33,296	33,285
800	33,296	33,272
1,000	33,306	33,304
2,000	33,332	33,333
4,000	33,337	33,337

PNG copies to `detailed/figures/`, exact plot-ready aggregates to `detailed/plot_data/`, and result-table rows to `detailed/tables/`. The script asserts the expected K2000 five-seed population counts and G-set 54-instance status total before plotting. The figure captions identify whether they show a five-seed endpoint summary, one coherent trajectory, fixed planted-instance means, or an adaptive attainment inventory. Those distinctions are part of the figure data, not small-print exceptions.

Appendix C: Complete numerical tables

1. Nested K2000 population sweep

Table VII prints the values underlying Fig. 2. The same five seed labels are used at each population size, and smaller populations are prefixes of the 8,192 initial fields. The target is 33,337 and the saved LQA mean and best are both 33,249.

TABLE VII. K2000 QeFEM endpoint cuts over five seeds at each nested population size. The best-gap column is relative to the stored 33,337 reference.

replicas	mean cut	best cut	best gap
128	33,252.0	33,255	82
512	33,254.6	33,255	82
1,024	33,260.8	33,274	63
2,048	33,272.6	33,293	44
4,096	33,277.8	33,293	44
8,192	33,281.2	33,293	44

2. All fixed Wishart files

TABLE VIII. Wishart results by requested α . Each method has five trial populations per fixed file. $\overline{\Delta E}$ is mean excess over the supplied planted energy; $\bar{\varepsilon}$ is mean relative energy error. Hits count trial populations with an exact endpoint.

α	mean ΔE		mean ε (%)		exact hits	
	QeFEM	LQA	QeFEM	LQA	QeFEM	LQA
0.1	0.18	0.59	0.716	2.368	0/5	0/5
0.2	0.79	1.53	1.586	3.059	0/5	0/5
0.3	1.93	3.04	2.564	4.036	0/5	0/5
0.4	3.70	5.08	3.687	5.057	0/5	0/5
0.5	6.04	7.36	4.822	5.879	0/5	0/5
0.6	8.98	11.11	5.983	7.408	0/5	0/5
0.7	12.54	15.09	7.174	8.630	0/5	0/5
0.8	16.66	19.30	8.360	9.681	0/5	0/5
0.9	13.09	24.53	5.825	10.920	2/5	0/5
1.0	0.00	30.95	0.000	12.373	5/5	0/5

3. All fixed Chook tile-planted files

TABLE IX. Chook results by requested tile-mixture parameter $p(C_3)$. Each method has ten trial populations per fixed file. Notation follows Table VIII.

$p(C_3)$	mean ΔE		mean ε (%)		exact hits	
	QeFEM	LQA	QeFEM	LQA	QeFEM	LQA
0.0	15.00	57.40	0.732	2.803	0/10	0/10
0.1	11.80	46.80	0.594	2.354	0/10	0/10
0.2	10.00	31.60	0.514	1.625	0/10	0/10
0.3	7.40	25.20	0.389	1.326	0/10	0/10
0.4	3.20	21.80	0.175	1.192	0/10	0/10
0.5	0.00	18.60	0.000	1.039	10/10	0/10
0.6	0.00	15.80	0.000	0.919	10/10	0/10
0.7	0.00	7.20	0.000	0.429	10/10	0/10
0.8	0.00	5.20	0.000	0.318	10/10	0/10
0.9	0.00	2.20	0.000	0.138	10/10	4/10
1.0	0.00	0.80	0.000	0.052	10/10	6/10

4. Every G1–G54 retained cut

The status labels below have strict provenance meanings. “Prior” means the supplied reference had already been attained before the later 31-graph search. “Fresh” means a new match was recovered by the frozen selected setting on fresh evaluation seeds. “Search” means a new match was seen during discovery or promotion but not recovered on its fresh seeds. “Open” means no saved result attained the supplied reference. These are evidence labels, not intrinsic classes of graph difficulty.

TABLE X. Complete retained G1–G54 results from the accumulated search. Targets are the supplied benchmark references, not certificates of global optimality. The two blocks continue the same table.

graph	ref.	best	gap	evidence	graph	ref.	best	gap	evidence
G1	11624	11624	0	prior	G28	3298	3298	0	search
G2	11620	11620	0	prior	G29	3405	3404	1	open
G3	11622	11622	0	prior	G30	3413	3412	1	open
G4	11646	11646	0	prior	G31	3310	3309	1	open
G5	11631	11631	0	prior	G32	1410	1406	4	open
G6	2178	2178	0	prior	G33	1382	1376	6	open
G7	2006	2006	0	prior	G34	1384	1378	6	open
G8	2005	2005	0	prior	G35	7687	7674	13	open
G9	2054	2054	0	prior	G36	7680	7664	16	open
G10	2000	2000	0	prior	G37	7691	7673	18	open
G11	564	564	0	prior	G38	7688	7674	14	open
G12	556	556	0	prior	G39	2408	2404	4	open
G13	582	582	0	prior	G40	2400	2398	2	open
G14	3064	3063	1	open	G41	2405	2402	3	open
G15	3050	3050	0	fresh	G42	2481	2473	8	open
G16	3052	3052	0	fresh	G43	6660	6660	0	prior
G17	3047	3047	0	search	G44	6650	6650	0	prior
G18	992	992	0	fresh	G45	6654	6654	0	prior
G19	906	906	0	fresh	G46	6649	6649	0	prior
G20	941	941	0	prior	G47	6657	6657	0	prior
G21	931	931	0	fresh	G48	6000	6000	0	prior
G22	13359	13359	0	fresh	G49	6000	6000	0	prior
G23	13344	13342	2	open	G50	5880	5880	0	prior
G24	13337	13337	0	fresh	G51	3848	3847	1	open
G25	13340	13340	0	fresh	G52	3851	3848	3	open
G26	13328	13326	2	open	G53	3850	3846	4	open
G27	3341	3341	0	prior	G54	3852	3848	4	open